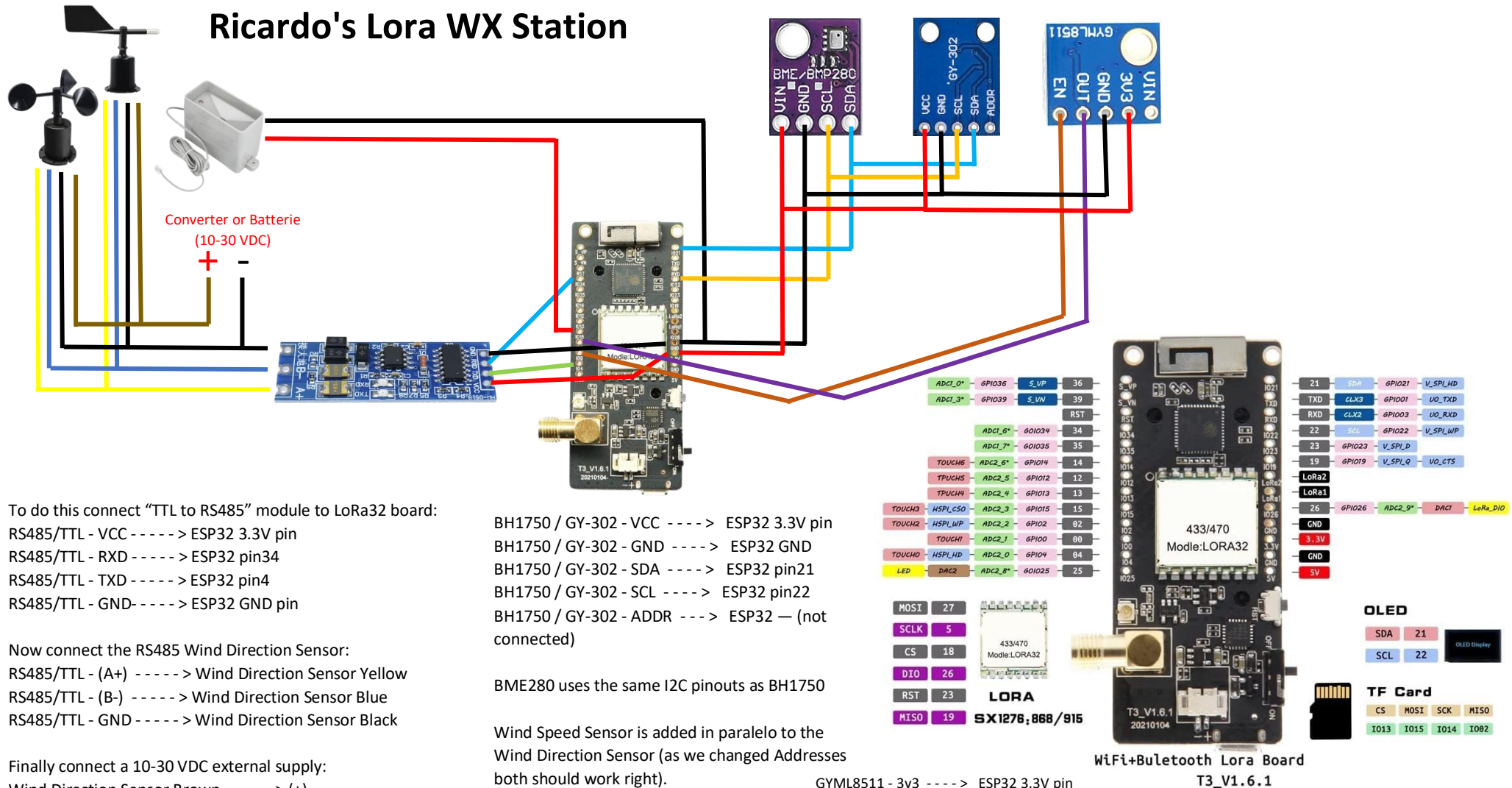


Ricardo's Lora WX Station



To do this connect “TTL to RS485” module to LoRa32 board:

```
RS485/TTL - VCC -----> ESP32 3.3V pin
RS485/TTL - RXD -----> ESP32 pin34
RS485/TTL - TXD -----> ESP32 pin4
RS485/TTL - GND-----> ESP32 GND pin
```

Now connect the RS485 Wind Direction Sensor:

RS485/TTL - (A+) -----> Wind Direction Sensor Yellow
RS485/TTL - (B-) -----> Wind Direction Sensor Blue
RS485/TTL - GND -----> Wind Direction Sensor Black

Finally connect a 10-30 VDC external supply:

Wind Direction Sensor Brown - - - - - > (+)
Wind Direction Sensor Black - - - - - > (-)

YES, all GND are connected.

```
BH1750 / GY-302 - VCC ----> ESP32 3.3V pin
BH1750 / GY-302 - GND ----> ESP32 GND
BH1750 / GY-302 - SDA ----> ESP32 pin21
BH1750 / GY-302 - SCL ----> ESP32 pin22
BH1750 / GY-302 - ADDR ---> ESP32 — (not
connected)
```

BME280 uses the same I2C pinouts as BH1750

Wind Speed Sensor is added in paralelo to the Wind Direction Sensor (as we changed Addresses both should work right).

Rain Sensor - GND - - - - > ESP32 GND
Rain Sensor - (+) - - - - > ESP32 pin15

```
GYML8511 - 3v3 ----> ESP32 3.3V pin
GYML8511 - GND ---> ESP32 GND
GYML8511 - OUT ---> ESP32 pin2
GYML8511 - EN ----> ESP32 pin0
GYML8511 - VIN ---> ESP32 — (not connected)
```